

stlToSolid — User Manual

Mesh in · solid out

Version 0.7.1 · November 2026

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2	Getting started	The screen layout, loading a file, units and scale, picking bodies and moving around the 3D view.
3	Mesh → Solid	Turns a machined or CAD-designed mesh into a clean STEP solid with real flat faces, cylinders and holes. Also gives you editable CadQuery and Fusion 360 scripts.
4	Sliced Loft	Slices a smooth, organic shape (a cap, handle or shell) along one axis and lofts the outlines into one smooth solid.
5	X-Ray	Turns slices of the mesh into real Fusion 360 sketches (lines, arcs and splines, not thousands of points) that you can select and extrude. Downloads a whole stack as one script, and can extrude them into solid slabs.
6	Blueprint	Reads a dimensioned drawing (a picture) into named parameters and sketch-and-extrude features, checks its own build against the drawing, and builds a parametric Fusion 360 script and a STEP. Parts the recipe cannot draw are written as a CadQuery script (STEP only).
7	Fusion 360 add-in: Mesh to Outline	The X-Ray and single-slice tracing inside Fusion itself: pick a mesh body and planes, get sketches (and optionally a solid) straight in your design. Also describes the STL to Solid add-in, not released yet.
8	Reading the report	What the result badge, the quality checks and the output table mean.
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12	Using stlToSolid from an AI assistant	Connect Claude, Cursor or another MCP client, so it can inspect and convert meshes and build parts from drawings for you.

1. Introduction

A mesh file (STL, OBJ, 3MF and so on) describes a part as thousands of tiny triangles. That is fine for a 3D printer, but a CAD program cannot work with it: you cannot sketch on a triangle, measure a hole or fillet an edge. Fusion 360's free tier can turn a mesh into a solid, but every triangle becomes its own face, so the "solid" is just as hard to model against.

stlToSolid rebuilds the part the way a designer would have drawn it: real flat faces, real cylinders and cones, real holes. You get a STEP file you can import into Fusion 360 (or any CAD program) and design around.

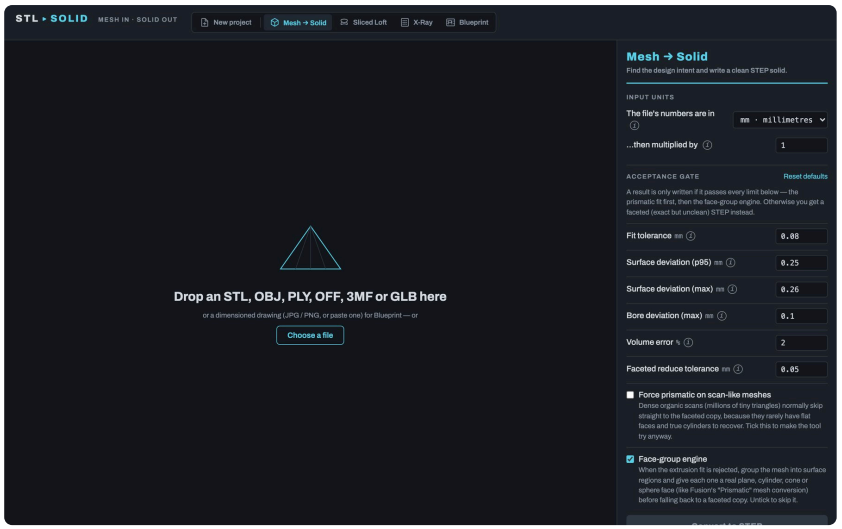
stlToSolid comes as a desktop app for Mac and Windows, as a Docker container for a home server or NAS (opened in a browser), and as an add-in that runs inside Fusion 360. The desktop app and the browser version are the same program with the same four tools. This manual covers all of them.

Which tool should I use?

Your part looks like…	Use	You get
A machined or CAD-designed part: brackets, plates, housings, holes, slots and steps	**Mesh → Solid**	A clean STEP with true planes, cylinders, cones, spheres and tori
A smooth, flowing shape that changes gradually along one direction: caps, handles, shells, bottles	**Sliced Loft**	One smooth lofted solid, plus a Fusion script that repeats the loft
A part you want to rebuild yourself in Fusion from cross-sections	**X-Ray**	Real, selectable Fusion sketches, one per slice, ready to extrude (optionally extruded into slabs for you)
No mesh at all, just a dimensioned drawing or datasheet picture	**Blueprint**	A parametric Fusion script (User Parameters, sketches, extrudes) and a STEP
A mesh that is already inserted in a Fusion design	**Mesh to Outline** (Fusion add-in)	The same X-Ray sketches, drawn straight into your design, with no file to download

2. Getting started

2.1 The screen



The start screen, before a file is loaded.

The window has three areas:

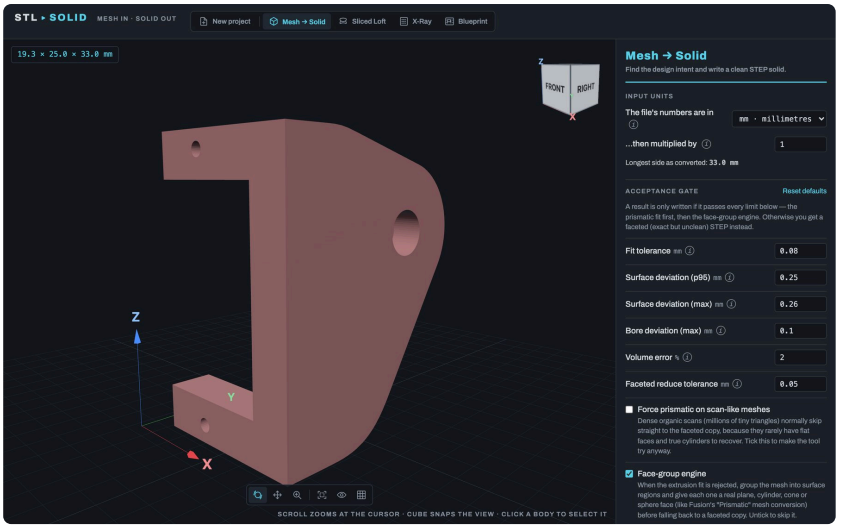
- **Top bar.** The **New project** button (start over with another file) and the four tool buttons: **Mesh → Solid**, **Sliced Loft**, **X-Ray** and **Blueprint**. The highlighted button is the active tool. Hover over a button for a one-line description.
- **Stage (left).** Drop a file here. Once loaded, it shows the part in 3D (or, for Blueprint, the drawing).
- **Rail (right).** The active tool's settings, its action button and, after a run, the report with the download buttons.

You can switch tools at any time. The loaded file stays, and each tool keeps its own settings.

2.2 Loading a file

Drag a file onto the stage, or click **Choose a file**. You can also paste a drawing image from the clipboard with **Cmd+V / Ctrl+V**.

- **Meshes:** `.stl`, `.obj`, `.ply`, `.off`, `.3mf`, `.glb`, `.gltf`. A mesh opens in the mesh tools. If Blueprint was active, the app switches to Mesh → Solid.
- **Drawings:** `.jpg`, `.png`, `.webp`. A drawing always opens in Blueprint.

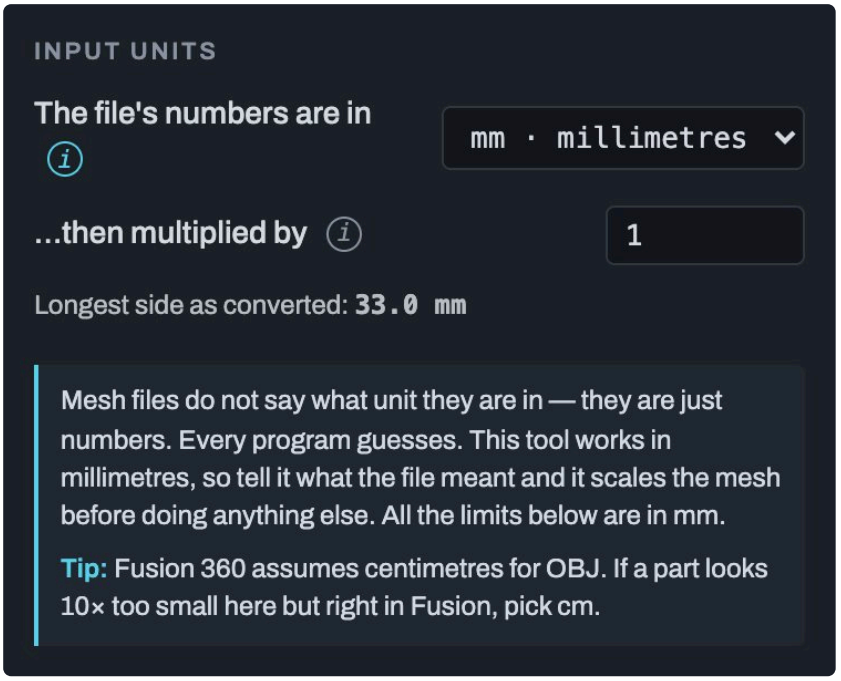


A servo bracket loaded. Its size is in the top-left corner of the stage.

The part's size appears in the top-left corner of the stage (for example `19.3 × 25.0 × 33.0 mm`). After a run, the **Input mesh** table in the report lists its triangle count, whether it is watertight (fully closed), how many bodies it holds, its bounding box, surface area and volume.

To start again with a different file, click **New project**.

2.3 Input units and scale



Input units. The small

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opens a plain-language explanation, shown here for the unit.

Mesh files do not record their units. They are just numbers. stlToSolid works in millimetres, so tell it what the file meant. These two settings are shared by Mesh → Solid, Sliced Loft and X-Ray.

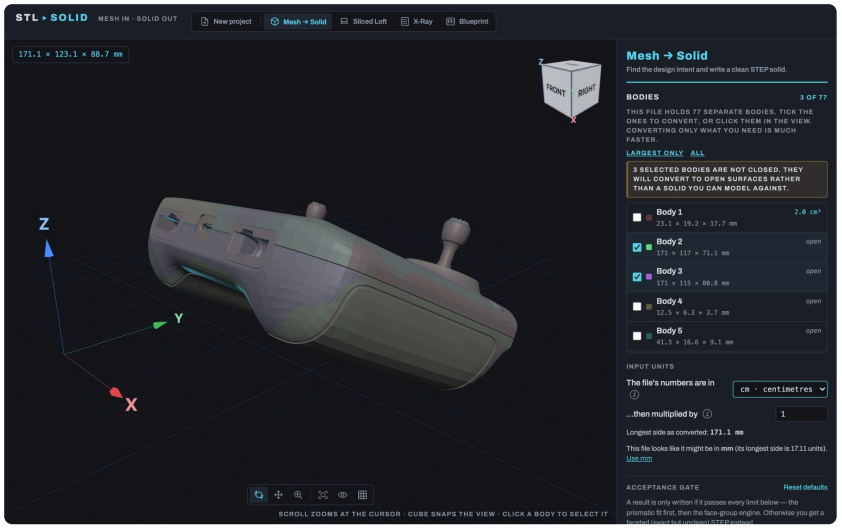
Setting	Default	What it does
The file's numbers are in	mm	The unit the file was saved in: mm, cm, inches, feet or metres. The mesh is scaled to millimetres before anything else happens.
…then multiplied by	1	An extra factor on top of the unit. Units can only make a part bigger (cm is ×10, inches ×25.4), so use this to shrink one. For example, `0.1` fixes a part designed in centimetres but saved as millimetres. Leave it at `1` unless the size looks wrong.

The line **Longest side as converted** shows the result, so you can spot a wrong scale before a long conversion, not after. If the part looks unusually large or small, a yellow **Check the size** note suggests a unit or factor and offers a one-click **Use …** button.

> **Tip:** Fusion 360 assumes centimetres for OBJ files. If a part looks 10× too small in stlToSolid but right in Fusion, pick **cm**.

2.4 Picking bodies

Some files hold several separate pieces ("bodies"). A scanned controller, for example, may have dozens. When that happens a **Bodies** panel appears at the top of the rail.

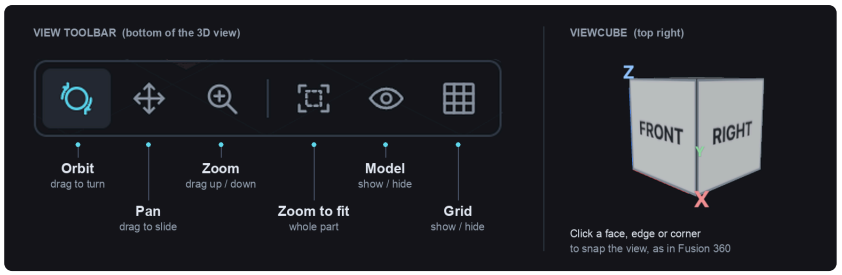


A controller with 77 bodies. Each body has its own colour in the view and in the list.

- Tick the bodies you want to convert, or **click them in the 3D view**. Hovering a row lights up that body.
- **Largest only** picks the main bodies. **All** ticks everything.
- Each row shows the body's size and volume. Bodies marked **open** are not closed. They convert to open surfaces rather than a solid, and a warning tells you when you have picked any.

Converting only what you need is much faster and uses much less memory.

2.5 Moving around the 3D view



The view toolbar sits at the bottom of the 3D view. The ViewCube sits in the top-right corner. The highlighted button is the active drag tool.

Control	What it does
Scroll wheel	Zooms in and out at the cursor.
ViewCube (top right)	Click a face, edge or corner to snap the view, as in Fusion 360.
Orbit (bottom toolbar)	Left-drag turns the view.
Pan	Left-drag slides the view.
Zoom	Left-drag up to zoom in, down to zoom out, at the point you press.
Zoom to fit	Frames the whole part.
Eye	Hides or shows the model. Slice traces stay visible.
Grid	Hides or shows the floor grid.

The coloured triad shows the axes: **X red, Y green, Z blue**, the same as Fusion 360.

3. Mesh → Solid

What it does: finds the design intent in a mesh (an extrusion direction, flat faces, cylinders, cones, spheres) and writes a clean STEP solid. It is the best tool for machined or CAD-designed parts.

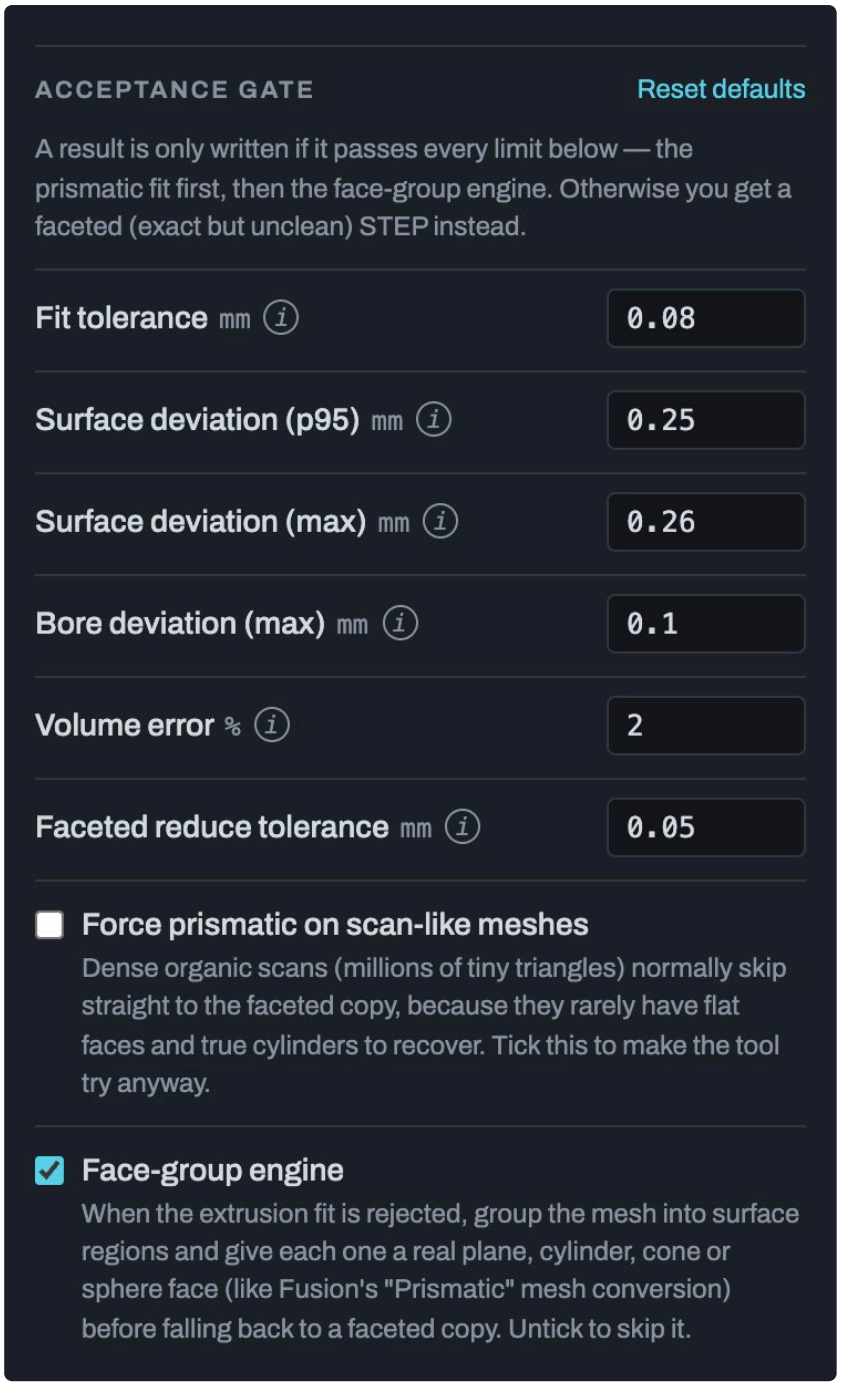
3.1 How it works

The tool tries three routes, best first, and keeps the first one that passes every quality check:

1. **Prismatic.** If the part is an extrusion (a sketch pulled straight), it redraws the outlines with lines and arcs and extrudes them. This is the cleanest result. It also gives you editable scripts.
2. **Face-group engine.** Otherwise it groups the mesh into surface regions and gives each one a real plane, cylinder, cone, sphere or torus face. This is the same idea as Fusion's paid "Prismatic" mesh conversion.
3. **Faceted.** If neither passes, you still get a valid, closed solid. Flat areas are merged into single faces and curved areas are simplified, but it is made of facets.

You always get a solid. The report tells you which route produced it.

3.2 Settings: the acceptance gate

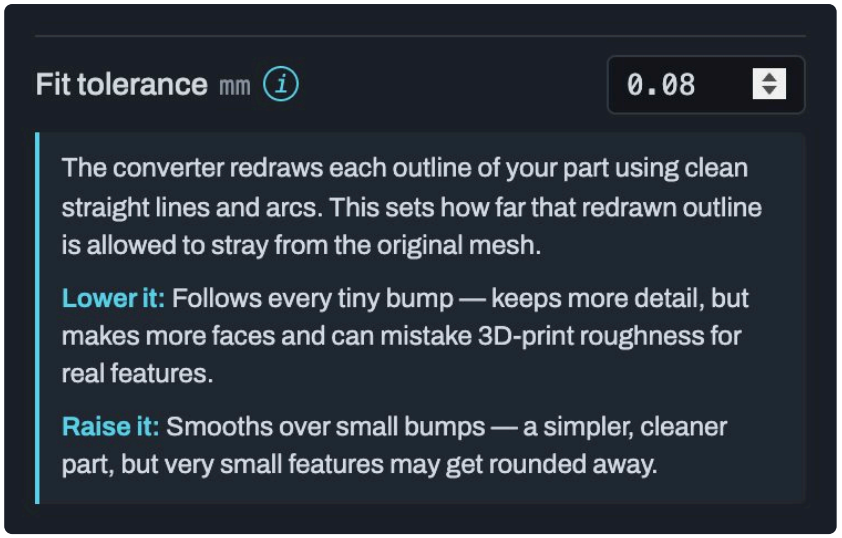


The acceptance gate and the two options. Changed values are highlighted.

Reset defaults

puts them all back.

A clean result is only written if it passes every limit. Click the small **i** next to any setting for a plain-language explanation of what raising or lowering it does.



The

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button opens the explanation, with what happens when you lower or raise the value.

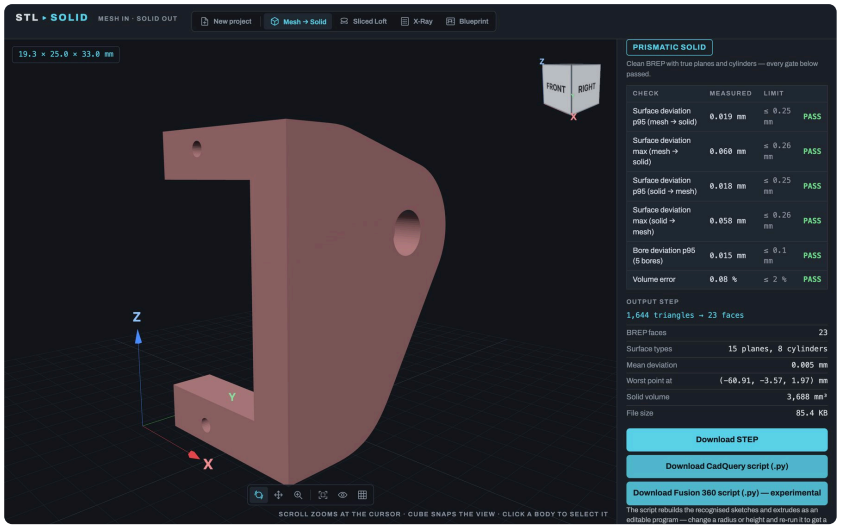
Setting	Default	What it means	Lower it	Raise it
Fit tolerance	0.08 mm	How far the redrawn lines and arcs may stray from the mesh.	Follows every tiny bump. Keeps more detail, but makes more faces and can mistake 3D-print roughness for real features.	Smooths over small bumps. A simpler part, but very small features may get rounded away.
Surface deviation (p95)	0.25 mm	95 % of the surface must be closer than this to the original. This is the main quality bar.	Demands a near-perfect match. More parts fall back to the faceted copy.	Accepts a rougher match. More parts convert cleanly, but may visibly differ.
Surface deviation (max)	0.26 mm	The limit for the single worst spot on the part. A taper, chamfer or rounded corner is usually what breaks it.	Even one slightly-off spot rejects the clean solid.	Lets one small area be off while the rest stays accurate.
Bore deviation (max)	0.10 mm	A stricter limit for round holes, so screws and pins still fit.	Holes must be almost exactly the right size.	Only sensible when the holes are cosmetic.
Volume error	2 %	The rebuilt part must hold about the same amount of material as the original. Catches a pocket that got filled in or a chunk that went missing.	Stricter overall shape check.	More forgiving. Rarely needs changing.
Faceted reduce tolerance	0.05 mm	Only used for a faceted result: how far simplified curved areas may move from the original (like Fusion's *Reduce by tolerance*). `0` keeps every triangle.	Keeps more triangles. Closer to the mesh.	Fewer triangles and a smaller STEP. Small details may be smoothed.

3.3 Settings: options

Option	Default	What it does
Force prismatic on scan-like meshes	off	Dense organic scans (millions of tiny triangles) normally go straight to a faceted copy, because they rarely have flat faces and true cylinders to recover. Tick this to make the tool try anyway.
Face-group engine	on	When the extrusion fit is rejected, try route 2 (fitted planes, cylinders, cones, spheres and tori) before falling back to a faceted copy. Untick to skip it.

3.4 Converting

Click **Convert to STEP**. The button shows *Converting…* (or *Waiting in queue…*) while it works, and a **Cancel** button appears. Your file and settings stay loaded if you cancel. Open **Pipeline log** under the report to watch the steps.



The servo bracket as a prismatic solid: 1,644 triangles became 23 faces (15 planes and 8 cylinders). Every check passed.

3.5 Downloads

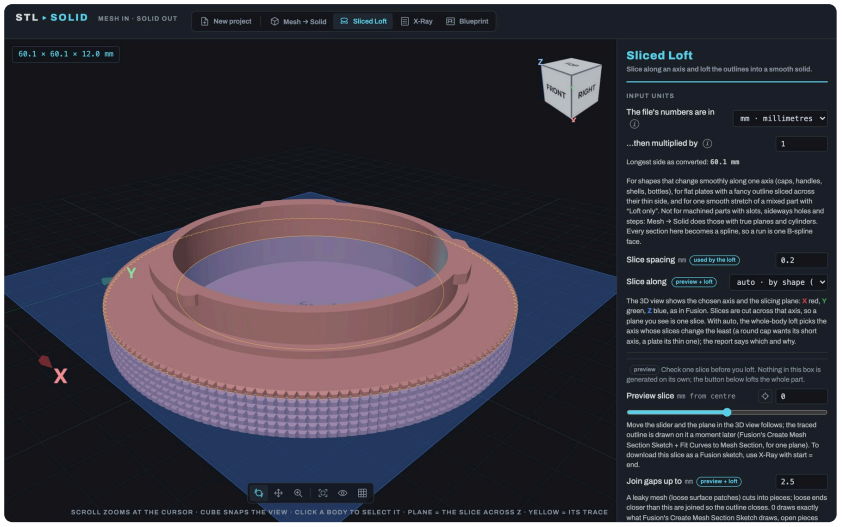
- **Download STEP**: the solid. Import it into Fusion 360 or any CAD program.
- **Download CadQuery script (.py)**: for prismatic results. It rebuilds the recognised sketches and extrudes as an editable program. Change a radius or height and re-run it to get a new STEP.
- **Download Fusion 360 script (.py) — experimental**: for prismatic results. It rebuilds the part as a parametric Fusion timeline.
- **Download Fusion 360 script (Boundary Fill)**: for face-group results (experimental). Fusion rebuilds every fitted surface and computes the edges itself, so the solid has clean edges. The label says whether a test run predicts it will build (*experimental*), probably fail (*likely to fail*), or could not be checked (*not checked*). The STEP is fine either way.

How to run a script in Fusion is in section 9.

4. Sliced Loft

What it does: cuts the part into thin slices along one axis, redraws each outline as a smooth curve and lofts the stack into one smooth solid. It is what you would do by hand in Fusion with *Create Mesh Section Sketch → Fit Curves to Mesh Section → Loft*, done automatically.

Use it for: shapes that change smoothly along one direction (lens caps, handles, shells, bottles), flat plates with a fancy outline (sliced across their thin side), and one smooth stretch of a mixed part (with **Loft only**). **Not for:** machined parts with slots, sideways holes and steps. A hole drilled across the slicing direction becomes a trough. Use Mesh → Solid for those parts.



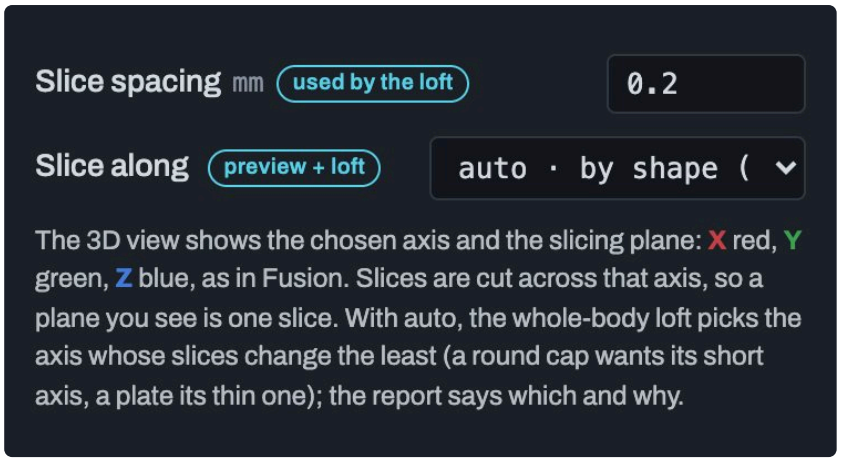
A camera body cap in Sliced Loft. The blue plane is the preview slice. The yellow line is the outline traced on it.

4.1 What the little tags mean

Some settings carry a small tag, so you can see which ones change the solid:

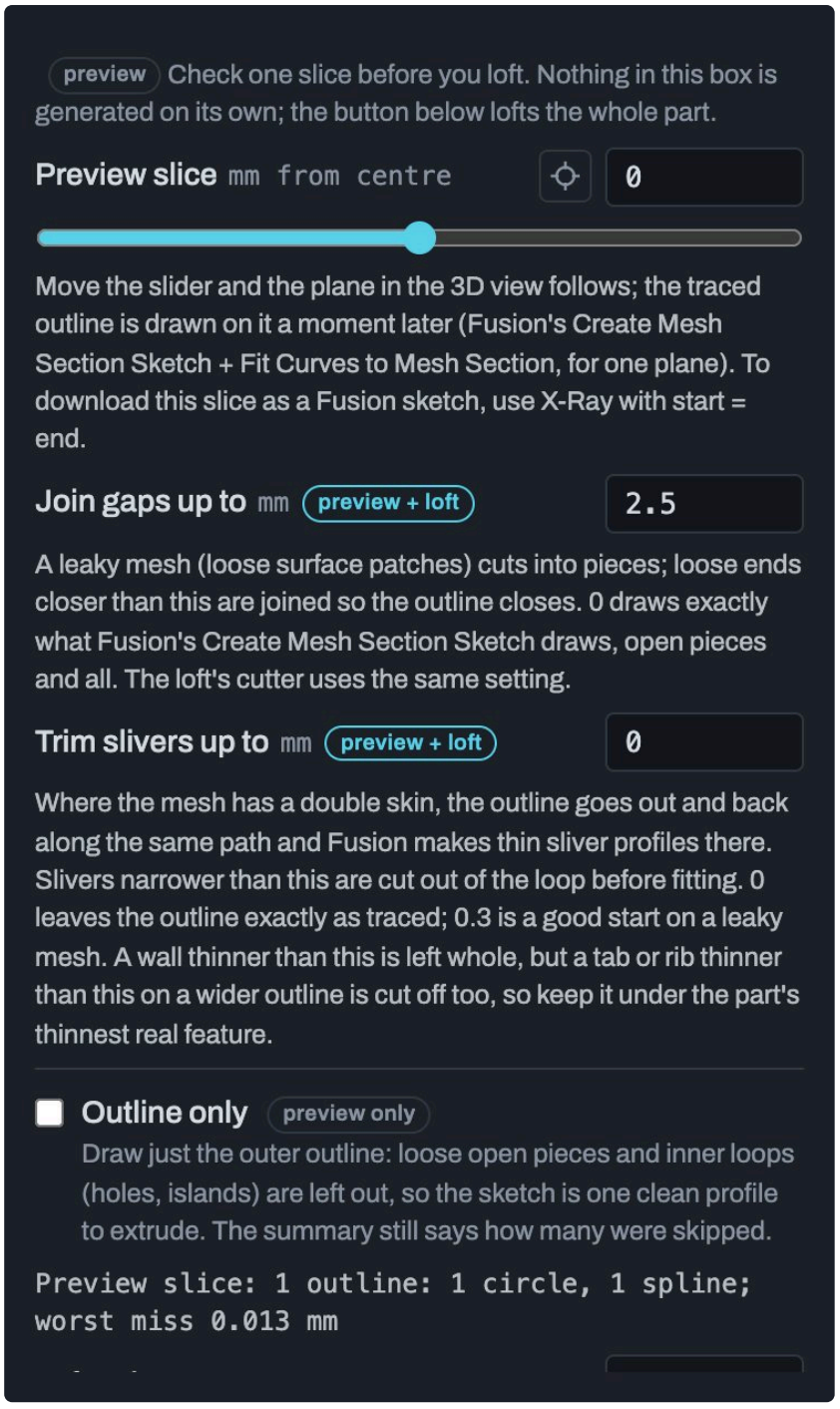
- **used by the loft**: changes the solid the Convert button makes.
- **preview + loft**: changes both the preview slice and the solid.
- **preview only**: changes only the preview slice. The solid ignores it.

4.2 Settings: spacing and axis



Setting	Default	What it does
Slice spacing	0.2 mm	Distance between slices. Smaller follows the shape more closely. The loft itself uses at most about 60 slices per run, but the checks still measure the result against every triangle of the mesh.
Slice along	auto	The axis to slice across. **auto · by shape** picks the axis whose slices change the least: a round cap wants its short axis, a plate its thin one. The list shows the part's length along each axis, and the report says which axis was picked and why.

4.3 Settings: the preview slice



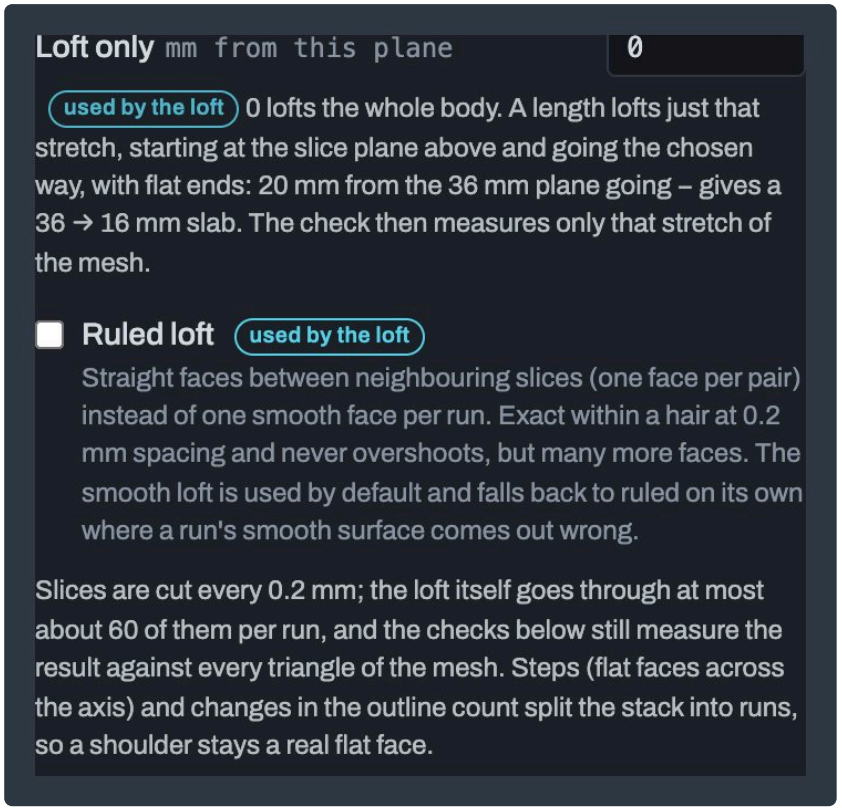
Everything in the preview box is for checking one slice before you loft. The summary line at the bottom says what was found.

Setting	Default	What it does
Preview slice	0 mm from centre	A slider and a number box that move one plane through the part. The traced outline appears on it a moment later. The **crosshair button** lets you click a face in the 3D view to put the plane there (Esc cancels). To download this one slice as a Fusion sketch, use X-Ray with start = end.
Join gaps up to	2.5 mm	A leaky mesh (loose surface patches) cuts into open pieces. Loose ends closer than this are joined so the outline closes. `0` draws exactly what Fusion's own mesh section draws, open pieces and all. The loft uses the same value.

Setting	Default	What it does
Trim slivers up to	0 mm	Where a mesh has a double skin, the outline goes out and back along the same path and makes thin slivers. Slivers narrower than this are cut out before fitting. Keep it at `0` for a clean, watertight mesh (a warning appears if you don't). `0.3` is a good start for a leaky one. Keep it below the part's thinnest real rib or tab.
Outline only	off	Draws just the outer outline and leaves out holes and loose pieces. Preview only.

The summary line reads like `Preview slice: 1 outline: 1 circle, 1 spline; worst miss 0.013 mm`: how many closed outlines were found, what they are made of, and how far the fitted curve is from the mesh at its worst point.

4.4 Settings: what to loft

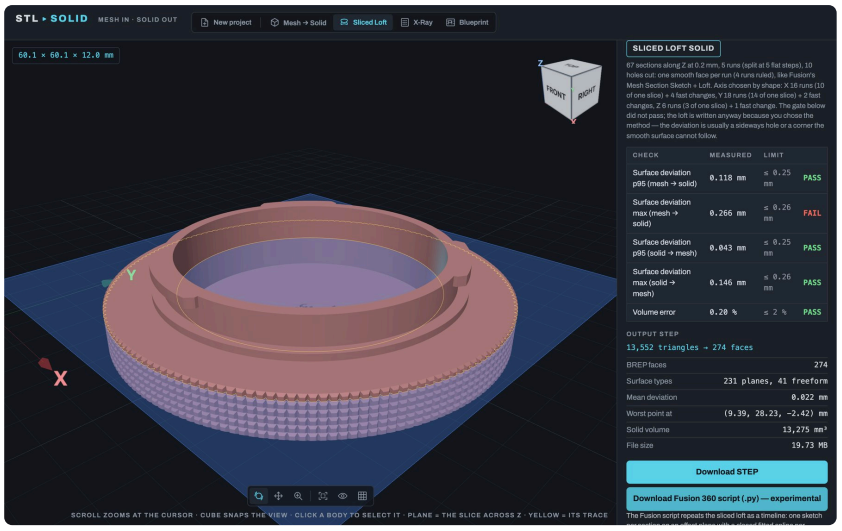


Setting	Default	What it does
Loft only … mm from this plane	0	`0` lofts the whole body. A length lofts just that stretch, starting at the preview plane, with flat ends. The preview slider is then labelled **Loft starts at**. For example, 20 mm from the 36 mm plane going – gives a 36 → 16 mm slab. The check then measures only that stretch of the mesh.
Going	–	Shown when Loft only is above 0. **+** goes along the axis arrow, **–** against it.
Ruled loft	off	Straight faces between neighbouring slices (one face per pair) instead of one smooth face per run. Exact within a hair at 0.2 mm spacing and never overshoots, but many more faces. The smooth loft switches to ruled by itself for any run whose smooth surface comes out wrong.

The acceptance gate appears below these settings too (see section 3.2). With Sliced Loft the limits are **measured and shown, not enforced**: you chose the method, so its solid is written whenever it can be built. Only a loft that cannot be built at all falls back to a faceted copy.

4.5 Converting and downloads

Click **Convert to STEP**. The stack is split into *runs* wherever a flat step crosses the axis or the number of outlines changes, so a shoulder stays a real flat face. Each run becomes one smooth face. Holes along the axis are lofted and cut.



The body cap as a sliced loft solid. The summary explains the sections, runs and chosen axis. Here one strict check (max deviation, 0.266 mm against 0.26 mm) is just over its limit. It is reported, but the solid is still written.

Downloads: Download STEP and **Download Fusion 360 script (.py) — experimental**. The script repeats the loft as a Fusion timeline: one sketch per section on an offset plane, then one Loft per run, with holes as Loft cuts.

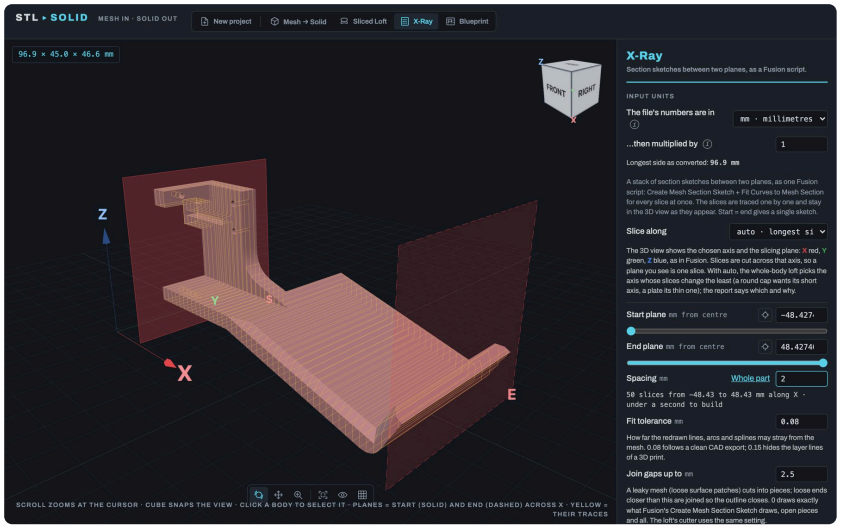
5. X-Ray

What it does: cuts the part at as many planes as you like and turns each cut into a **real Fusion 360 sketch**: clean lines, arcs and splines that join into closed profiles. You can click a profile and extrude it, offset it, or dimension against it, like a sketch you drew yourself.

Why this matters. Fusion's own *Create Mesh Section Sketch* gives you an outline made of hundreds or thousands of separate points. You cannot select it as a profile or extrude it, so in practice you have to draw a new sketch around the shape by hand. X-Ray does that redrawing for you, for every slice at once.

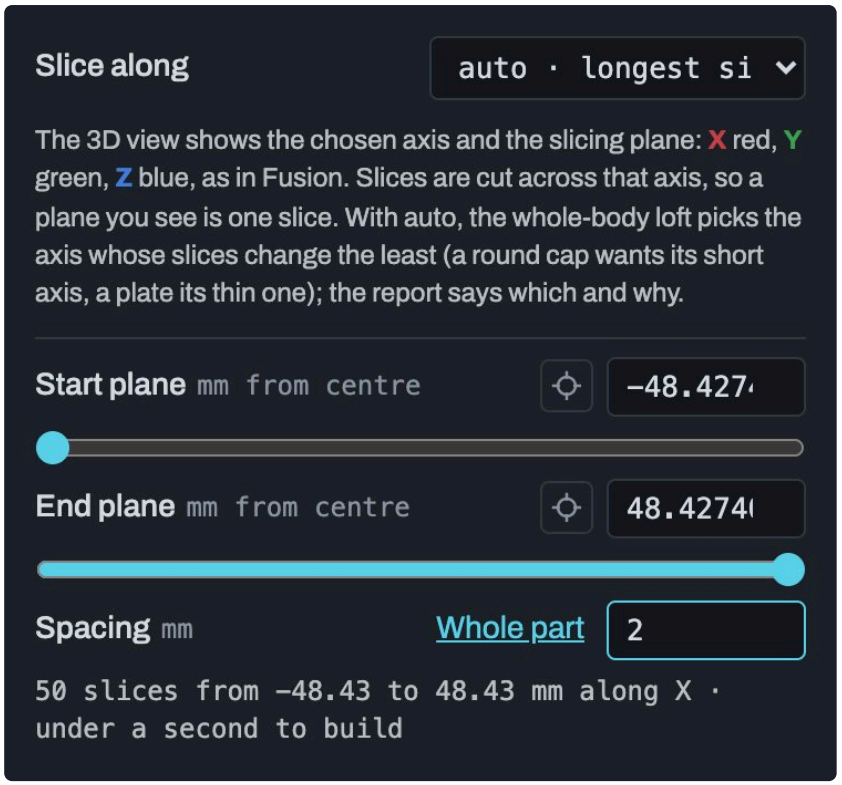
	Fusion mesh section sketch	X-Ray sketch
Made of	Hundreds or thousands of points	A few lines, arcs and splines
Closed profile you can select	No	Yes
Extrude, offset, dimension	Only after you trace it by hand	Straight away
Many slices	One plane at a time	A whole stack in one script

X-Ray does not make a STEP. It downloads one Fusion 360 script that draws the sketches, each on its own plane, so you can model in Fusion yourself. It can also extrude every slice into a solid slab for you. If the mesh is already in a Fusion design, the Mesh to Outline add-in does the same thing without a download.



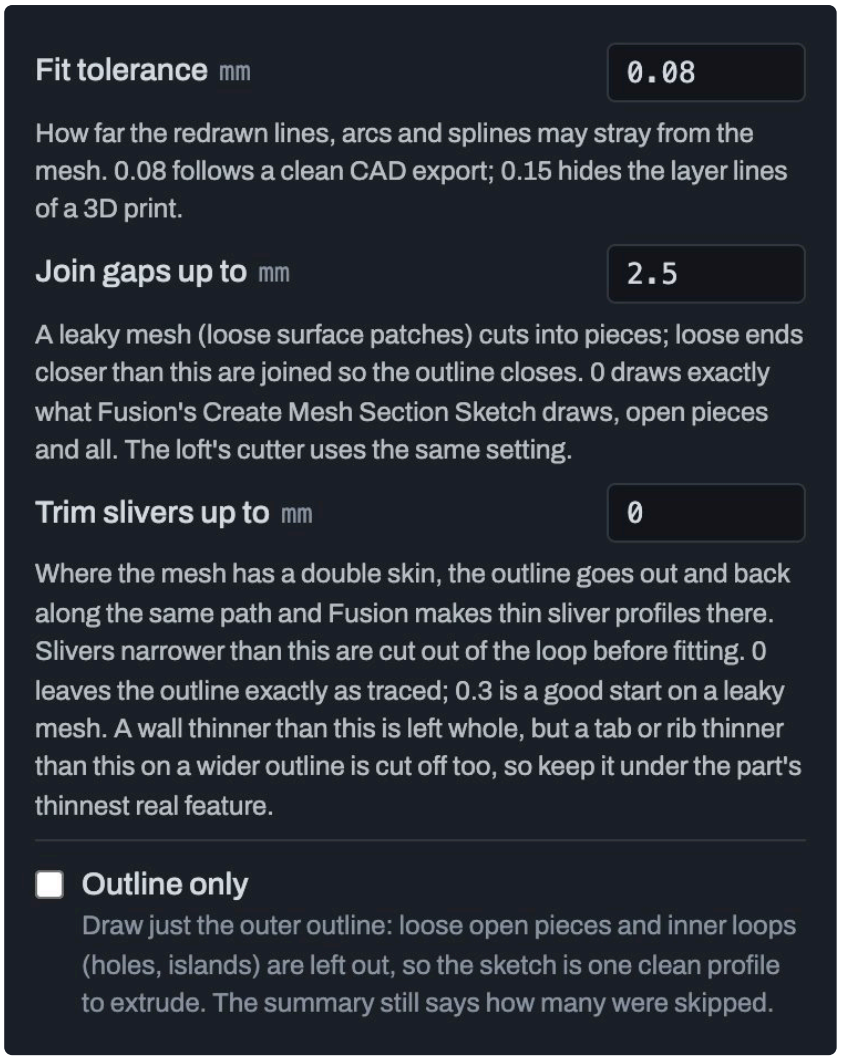
X-Ray on a frame, sliced along X every 2 mm over the whole part (50 slices). S is the start plane (solid), E the end plane (dashed). Each yellow outline is one traced slice, ready to become a Fusion sketch.

5.1 Settings: where to slice



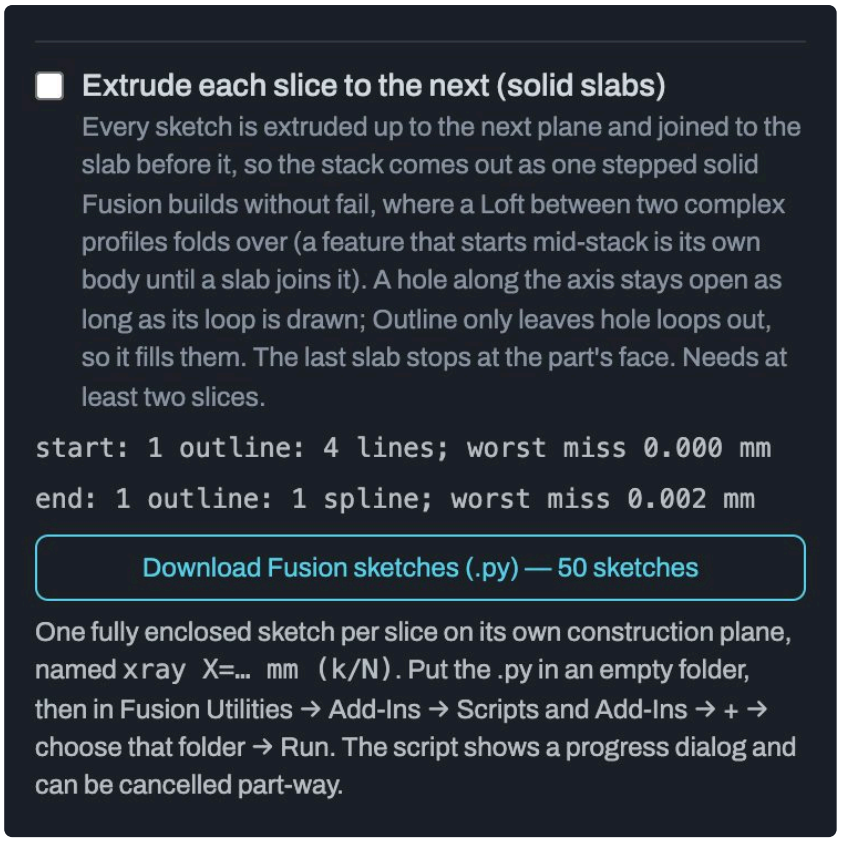
Setting	Default	What it does
Slice along	auto	The axis to slice across. In X-Ray, **auto** uses the longest side.
Start plane / End plane	0 mm from centre	Where the stack begins and ends, in mm from the part's centre. Each has a slider, a number box and a **crosshair button**: press it, then click a face in the 3D view to put the plane there. Start = end gives a single sketch. Start past end is fine.
Spacing	0.2 mm	Distance between slices. **Whole part** moves the planes to both ends of the part. The line below shows the slice count and an estimated build time.

5.2 Settings: how each slice is traced



Setting	Default	What it does
Fit tolerance	0.08 mm	How far the fitted lines, arcs and splines may stray from the mesh. `0.08` follows a clean CAD export. `0.15` hides the layer lines of a 3D print.
Join gaps up to	2.5 mm	As in Sliced Loft (section 4.3). X-Ray keeps its own value.
Trim slivers up to	0 mm	As in Sliced Loft. X-Ray keeps its own value, so a trim set for a leaky loft does not eat the thin walls of the next part. If slivers were trimmed anywhere in the stack, a note says in how many slices.
Outline only	off	Just the outer outline of each slice: holes, islands and loose pieces are left out. About five times faster on organic shapes.
Tidy corners (tangent fillets, sharp creases)	off	Fillets are solved to meet their walls tangent and corners meet exactly. In the Fusion script the curves of each loop share their end points and every fillet gets a tangent constraint, so the sketch behaves like one you drew. The summary says how many junctions are tangent. Off draws the plain fit.
Sharpen rounds below	0.08 mm	Shown with Tidy corners. A round between two walls that stays within this distance of their corner is drawn as the sharp corner, like Fusion's Fit Curves at the same tolerance. `0.08` turns rounds up to about 0.25 mm sharp. `0` keeps every round. The summary says how many were sharpened.

5.3 Settings: slabs and the download



The start and end summaries say what the two end slices hold. The download button shows how many sketches the script will draw.

Setting	Default	What it does
Extrude each slice to the next (solid slabs)	off	The script also extrudes every sketch up to the next plane and joins the slabs, giving one stepped solid that Fusion always builds. (A Loft between two complex profiles often folds over.) Holes along the axis stay open as long as their loops are drawn. Outline only leaves hole loops out, so it fills them. Needs at least two slices.
Add 2 rails (90° sections)	off	Two more cuts, along the stack through the part's centre, shown in cyan in the view. See *Rails and loft* below. Needs at least two slices.
Loft profiles + rails (new body)	off	Shown with rails. The script lofts one new body through the slices along the rails. Not with Extrude each slice.
Cut holes into the loft	on	Shown with the loft. Every hole that runs through two or more slices is cut out of the lofted body.

The slices are traced one by one and stay in the view as they appear. Click **stop** to halt the tracing and **trace the rest** to carry on.

Click **Download Fusion sketches (.py)**. The server fits every slice (*Working…* with a spinner) and then hands you the script. Each slice becomes a fully enclosed sketch on its own construction plane, named like `xray X=−18.50 mm (3/24)`. In Fusion, the script shows a progress dialog and can be cancelled part-way.

Limits: at most 500 slices and about 4 minutes of fitting per download. If you go over, raise the spacing, narrow the range, or tick **Outline only**.

Rails and loft

A Loft in Fusion follows the slices much better with rails, but Fusion refuses a rail that does not touch every profile exactly. The raw mesh cut misses the fitted outlines by up to the fit tolerance, so the rails here are pinned: where a rail crosses a slice, its point is moved onto that slice's fitted outline (onto the line or arc itself, and into a spline's fit points so the spline passes through it). The script writes every number at 6 decimals when there are rails, so they still touch after rounding.

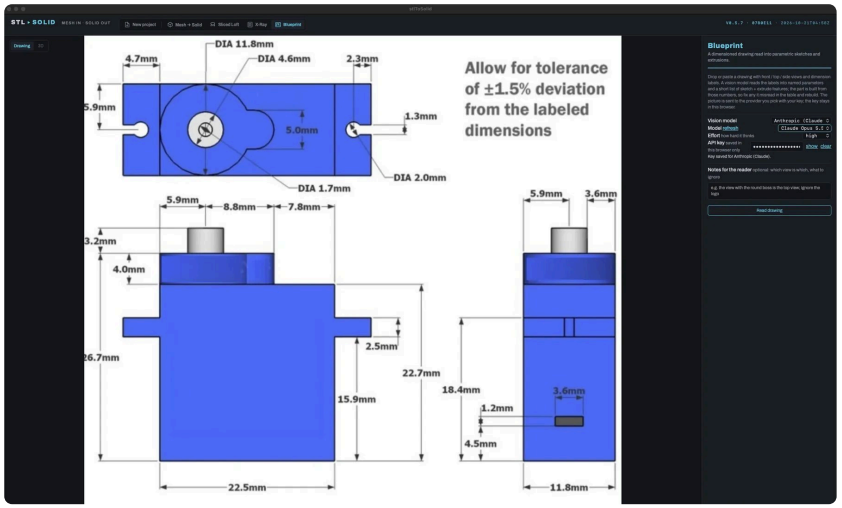
- The two rail planes go through the part's centre and contain the stack's axis. Each cuts the part's outer wall twice, so a simple part gets 4 rails.
- A rail is *complete* when it crosses every slice once, close to its outline. Others (a slot wall, a rail that misses a slice) are drawn but no loft uses them. The closing message lists them.
- The rails and the loft follow the outer outline. On a hollow part the cuts along the inner walls are drawn too, but marked as not touching every slice.
- Fusion's Loft fills holes: it uses each slice's outer outline only (it shows the hole outlines in red). With **Cut holes into the loft** the script finds every hole that runs on through two or more slices (same place, similar size) and cuts it out of the lofted body with a loft of its own, so a hole that changes shape along the stack follows it. The closing message says how many were cut. With **Outline only** there are no holes to cut.
- The loft is best effort. If Fusion refuses the rails it tries without them. If it refuses the loft too, the sketches and rails stay and the message says why. It is skipped past 60 slices: use a bigger spacing for a loft (5 mm on a hand-held part is a good start).

6. Blueprint

What it does: starts from a picture instead of a mesh. Drop or paste a dimensioned drawing (front, top and side views with their labels, like a datasheet). A vision AI model of your choice reads the labels into a small **recipe**: named parameters in millimetres and a short list of sketch + extrude features. The part is built from those numbers, so you can fix any it misread and rebuild. You get a **parametric Fusion 360 script**, a STEP and a 3D preview.

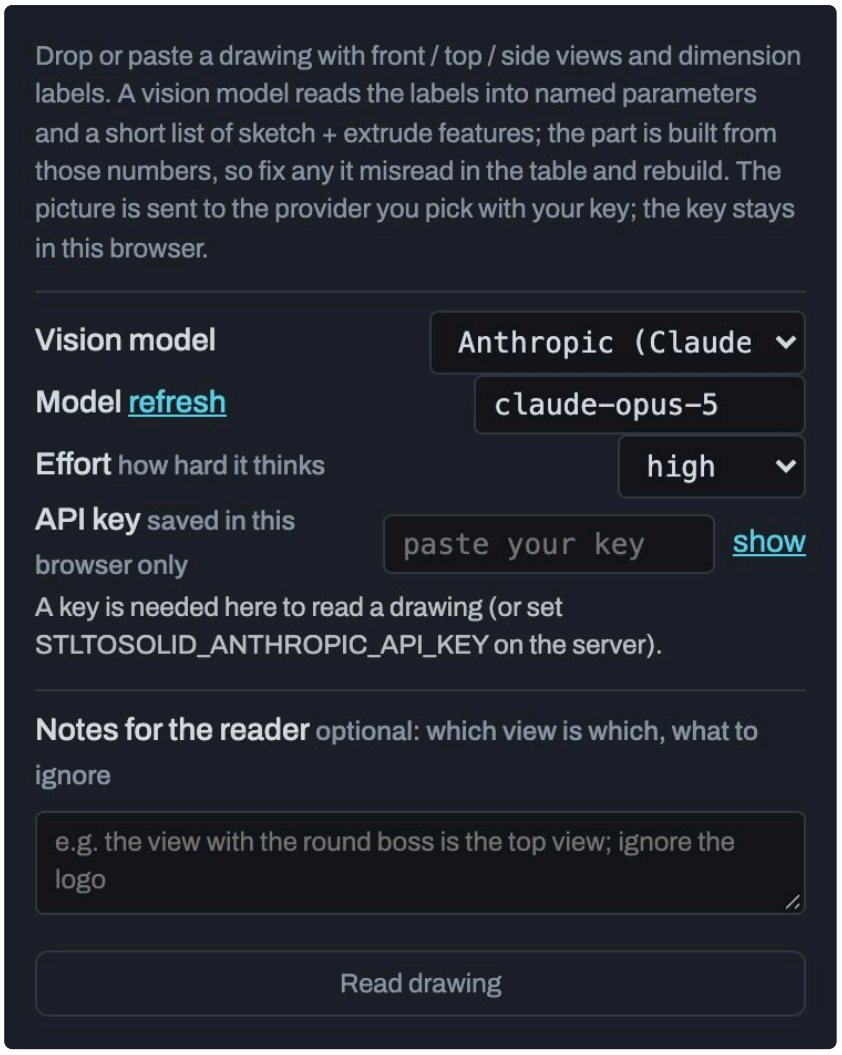
Two things happen on top of that read (both new in 0.5.8):

- **Self-check.** After the read, the part is built and drawn from four sides (front, top, right, isometric). The pictures go back to the model next to the drawing with the question "what differs?", and it corrects its answer. Up to three such rounds; each is one more model call. Off unless you turn it on (see **Self-check** below).
- **Code mode.** The recipe language only knows flat sketches extruded straight. A bent bracket, a blended web, a boss square to a sloped arm or an edge fillet cannot be said in it. For such a part the model writes a short **CadQuery script** instead, which runs on the server under an allow-list. You get a STEP, the 3D preview and the script itself (you can edit a number and run it again), but no Fusion script: code is not a list of sketches.



A micro-servo datasheet drawing loaded in Blueprint, ready to read.

6.1 Settings: reading the drawing



Setting	Default	What it does
Vision model	Anthropic (Claude)	The provider that reads the drawing: Anthropic (Claude), OpenAI, DeepSeek, or *Custom* (any OpenAI-compatible server, such as Ollama running a vision model on your own computer, or OpenRouter).
Model	the provider's default	Choose from the list, click **refresh** to ask the provider again, or pick *type a name…* to enter one.
Effort	high	How hard the model thinks. *high* is the most careful, slowest and most expensive. *medium* takes about half the time. *low* is a quick look.
How to build	auto	*auto*: the model looks at the drawing first (one quick, cheap call) and picks a recipe when every feature is a straight extrude, code otherwise. *recipe*: always the parametric table and Fusion script. *code*: always a CadQuery script (any shape, STEP only). Greyed out when the server has code switched off (`STLTOSOLID_ALLOW_CODE=0`).
Self-check	off	Look-and-fix rounds, 0 (*off*) to 3. Each round builds the part, draws it from four sides and asks the model to compare the pictures with the drawing, then corrects its answer. Rounds run at *medium* effort when the read ran at *high*. Off by default: a round can also talk the model out of a right reading, so try 1 or 2 on a part and look at the pictures.

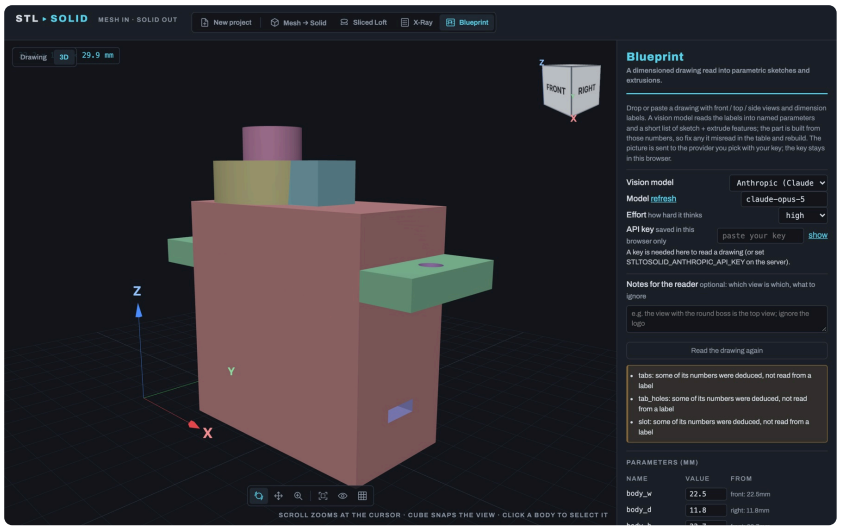
Setting	Default	What it does
Base URL	—	*Custom* only: the server's address, for example `http://localhost:11434/v1` for Ollama.
API key	empty	Your key for that provider. **show** reveals it, **clear** removes it. It is saved in this browser (or app) only and sent with each read. The server never stores it. Local servers usually need no key.
Notes for the reader	empty	Optional hints, for example *"the view with the round boss is the top view; ignore the logo"*.

Click **Read drawing**. A **Cancel** button appears while it reads, and a line under the button says what is happening (*reading the drawing into a recipe…*, *check 1 of 2: comparing the views with the drawing…*).

> **Privacy:** the drawing is sent to the provider you pick, with your key, and nowhere else. Choose a local model if the drawing must stay on your computer. In code mode the model's script runs on the server that hosts the app (your computer for the desktop app). It is checked before it runs: only `cadquery` and `math` may be imported, there is no file access, no `while`, and a run is stopped after two minutes.

6.2 Checking and editing the recipe

When the read finishes, the part is built and the stage switches to the **3D** tab. Use the **Drawing | 3D** tabs to switch between the picture and the model.



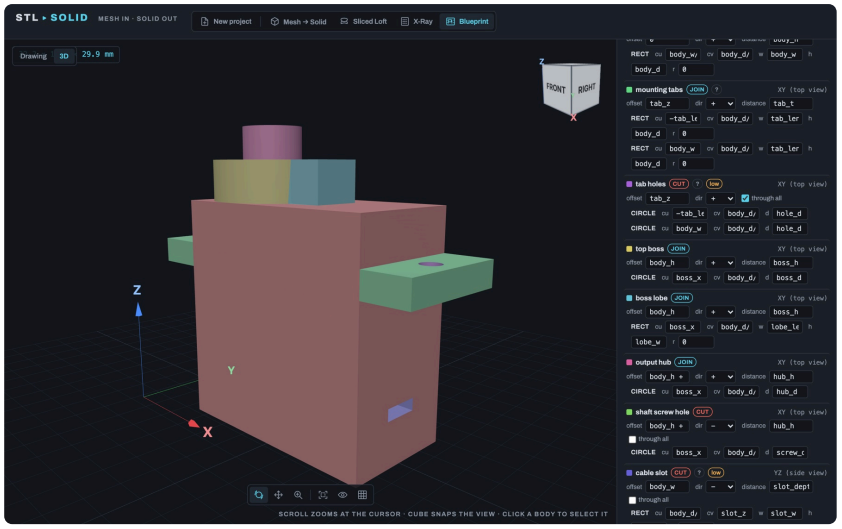
The built servo. Each feature has its own colour.

Before anything is built, the recipe goes through automatic checks: every value must resolve, shapes must have a size, cuts must remove something, and the overall size must match the drawing. Problems and guesses are listed in a yellow box.

Parameters (mm). Every number, with **From** saying where it came from (for example `front: 22.5mm`). A **?** marks a value the model had to deduce rather than read from a label. Check those first.

Features. Each feature has a colour swatch that matches the 3D view. Hover a feature to light it up. For each feature you can see and edit:

- the operation (**new body**, **join** or **cut**) and its sketch plane (XY = top view, XZ = front, YZ = right side),
- **offset**, direction (**+**, **–** or **sym**) and **distance**, or **through all** for cuts,
- its shapes (rectangle, circle, slot, polygon) with their positions and sizes (`cu` / `cv` = centre, `w` / `h` = width and height, `r` = corner radius, `d` = diameter). A **low** badge marks a feature the model was unsure of.



The feature list. Values can be numbers or expressions such as `body_w/2`.

Any value can be a number or an expression over the parameters, like `body_w/2`. Change a value and the 3D view redraws within about a second (*Redrawing object…*). Click **Revert** to undo your edits.

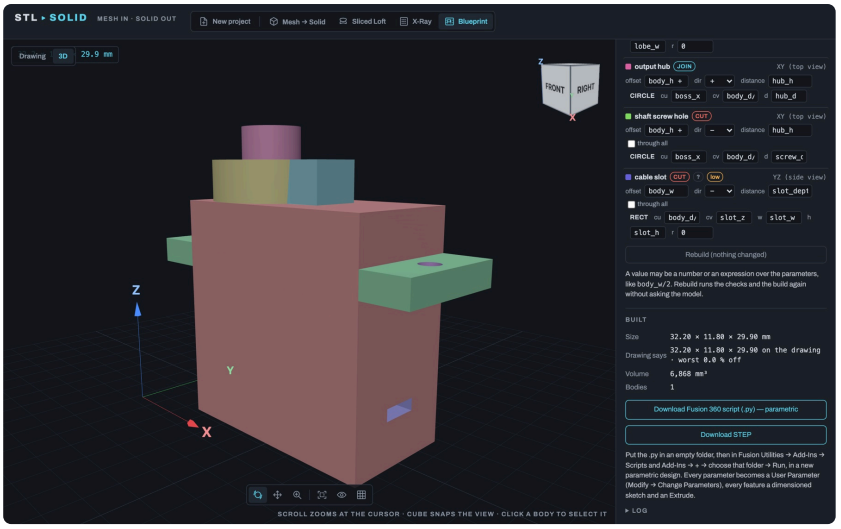
If the model notes that the recipe could not draw part of the drawing (a fillet, a bend, a tilted boss), a button **Read again as code** appears under the warnings; it reads the drawing again in code mode.

Self-check. Below the recipe (or the script), a *Self-check* list says what each round found: *check 1: 3 discrepancies fixed*, with the model's own words for each (*boss: sits upright, drawing shows it square to the sloped arm → tilt the boss plane 15°*), then *check 2: the build matches the drawing*. A correction that fails more checks than the answer before it is not taken.

Code mode. Instead of the table you see the **Script**: the model's named numbers at the top (with where each came from), the CadQuery code, and its notes on how it read the sheet (which view is which, how an ambiguous dimension was taken). Edit the script and click **Run the script again (STEP)**: no model call, the allow-list check first, then the build. **Revert** restores the model's version.

6.3 Rebuilding and downloading

Click **Rebuild with these values (STEP + Fusion script)**. This runs the checks and the build again **without** asking the AI model, so it costs nothing.



The Built table compares the model's size with the size stated on the drawing.

The **Built** table shows the size, what the drawing says (and how far off the worst dimension is), the volume, the number of bodies, and which model read the drawing, in which mode, with how many calls. Under it are the four pictures the model compared with the drawing (click one to open it full size). Then:

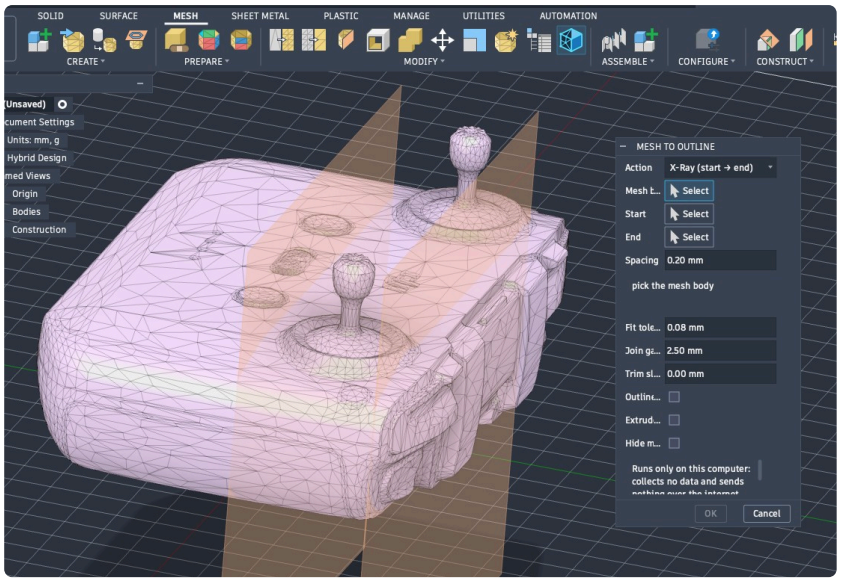
- **Download Fusion 360 script (.py) — parametric.** Every parameter becomes a Fusion **User Parameter** (*Modify → Change Parameters*), and every feature becomes a dimensioned sketch and an Extrude. Change `body_h` in Fusion and the parts on top move with it. (Recipe mode only: a part written as code has no Fusion script.)
- **Download STEP.** The same part as a plain solid.

Click **Read the drawing again** to start over with a fresh read.

7. Fusion 360 add-in: Mesh to Outline

What it does: the X-Ray and single-slice tracing, running inside Fusion 360 itself. Pick a mesh body and a plane, and it draws the section as a normal sketch of lines, arcs, circles and fitted splines. Or pick a start and an end and it draws a whole stack, and can extrude it into one solid. It is Fusion's *Create Mesh Section Sketch* followed by *Fit Curves to Mesh Section*, in one button, for as many planes as you like.

It runs on Fusion's own Python. Nothing is uploaded, and no server or app is needed.

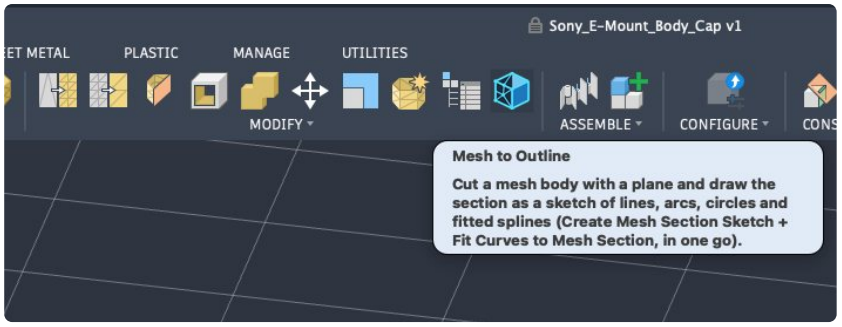


Mesh to Outline on a scanned controller in Fusion 360.

7.1 Installing

1. Download the zip for your computer from the latest release and unzip it.
2. Copy the `MeshSectionSketch` folder into Fusion's add-ins folder. Do not rename it. - **Mac:** `~/Library/Application Support/Autodesk/Autodesk Fusion 360/API/AddIns/` (the `Library` folder is hidden: in Finder press **Cmd+Shift+G** and paste the path) - **Windows:** `%APPDATA%\Autodesk\Autodesk Fusion 360\API\AddIns\`
3. In Fusion: **Utilities** → **Add-Ins** → **Scripts and Add-Ins** → **Add-Ins** tab, select **MeshSectionSketch** and click **Run**. It starts with Fusion from then on.

It needs a current Fusion (with its built-in Python 3.14) on macOS 15 or newer, or Windows 11. numpy comes inside the add-in. To update, stop the add-in, replace the folder with the new one and run it again: there is no auto-update.



The button sits on the

Mesh

tab, in the

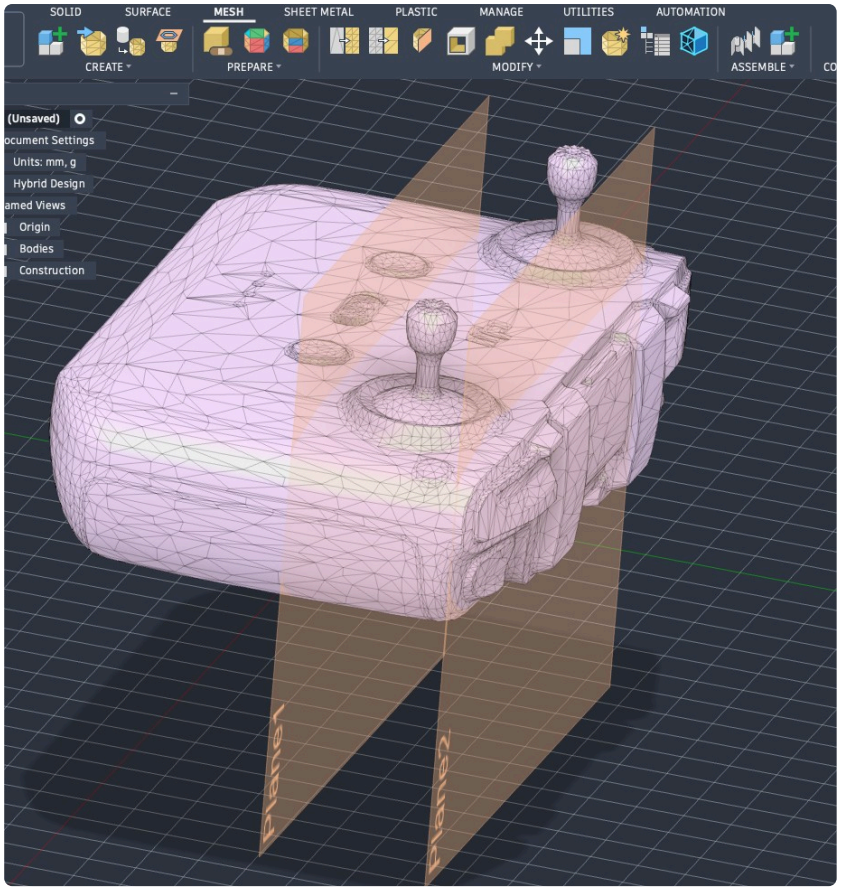
Modify

panel, next to Convert Mesh.

If the Mesh panel is not found, the button goes under **Utilities** → **Add-Ins** instead. The Text Commands palette says where.

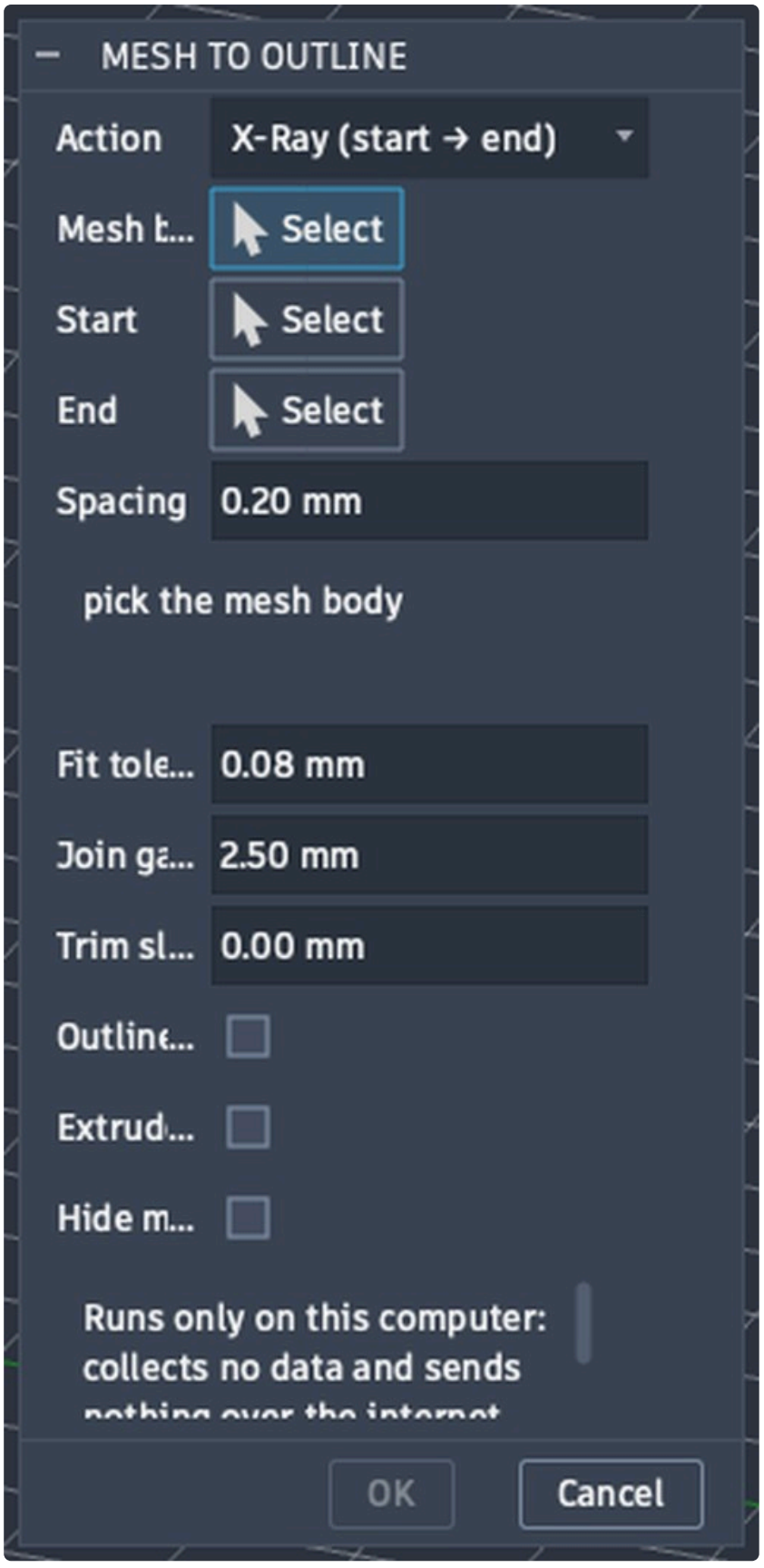
7.2 Before you start

1. **Insert** → **Insert Mesh**, and set the units to mm.
2. For a single section or a stack: make a construction plane where you want the cut (**Construct** → **Offset Plane**), or use any flat face. For an X-Ray: two parallel planes, or one plane and a click on the mesh.
3. **Mesh** tab → **Modify** → **Mesh to Outline**.



Ready for an X-Ray: the mesh and two planes, before the dialog is open.

7.3 The settings



The dialog in X-Ray mode with nothing picked yet, showing the defaults. Fusion cuts the labels short. The full names are in the table below.

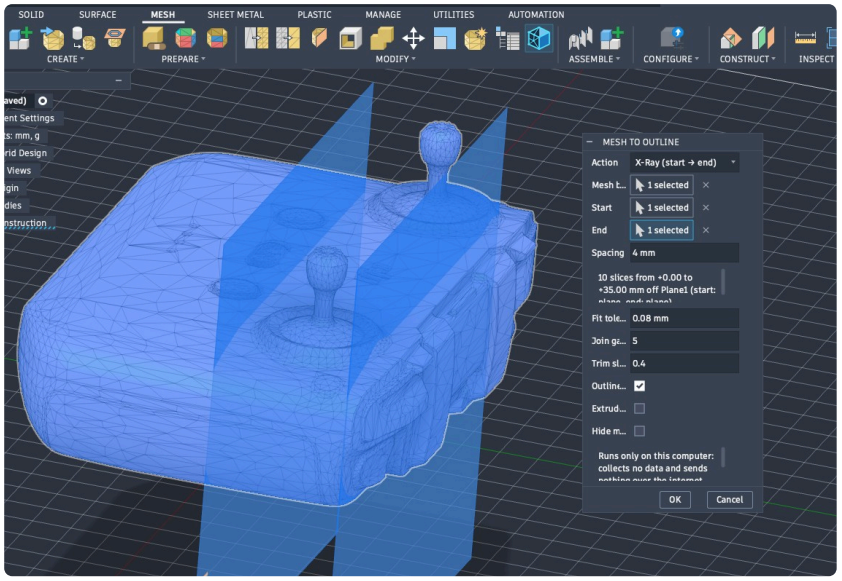
Setting	Default	What it does
Action	Single slice	Choose this first: it decides which picks the dialog asks for. **Single slice**: a single sketch on the plane. **Stack (count + direction)**: several sketches on offset planes. **X-Ray (start → end)**: a stack between two picks, like the app's X-Ray. Switching clears the picks the new action does not use.
Mesh body	—	The mesh body to cut.
Plane	—	Single slice and Stack: a construction plane or a flat face to cut on.
Start / End	—	X-Ray only: where the stack starts and ends. Pick a construction plane or a flat face, or click on the mesh. The slices run parallel to the Start plane, or the End plane if Start is a mesh click, so at least one must be a plane or a flat face of a solid, and two planes must be parallel. A click on a flat mesh face across the slice direction snaps to that face (just inside the material, so the cut catches it). A click anywhere else takes the height of the spot clicked. The end is always included, and start past end is fine.
Spacing	0.2 mm	Stack and X-Ray: the distance between sketches.
Sections (count)	10	Stack only: how many sketches, 2 to 400.
Direction	Along the plane normal (+)	Stack only: along the plane's normal (+), against it (–), or both sides of it.
(info line)	—	X-Ray only: how many slices your picks and spacing make, and between which heights.
Fit tolerance	0.08 mm	How far a fitted curve may sit off the raw section.
Join gaps up to	2.5 mm	Loose ends closer than this are joined, so an outline broken only by hairline cracks between surface patches comes back as one closed profile. `0` draws exactly what Fusion's own mesh section draws. Raise it if the outline still comes back in pieces. Lower it if two unrelated pieces got joined.
Trim slivers up to	0 mm	Where the mesh has a double skin, the outline goes out and back along the same path and Fusion makes thin sliver profiles. Slivers narrower than this are cut out before fitting. `0.3` is a good start on a leaky mesh. `0` leaves the outline as traced.
Outline only	off	Just the outer outline. Loose open pieces and inner loops (holes, islands) are skipped, so each sketch is one clean profile to extrude. The summary still says how many were skipped.
Tidy corners	off	Fillets meet their walls tangent and corners meet exactly; the curves of each loop share their end points and every fillet gets a tangent constraint. The closing message says how many constraints were added (and how many Fusion refused).
Sharpen rounds below	0.08 mm	Shown with Tidy corners. Rounds that stay within this of their corner are drawn as the sharp corner. `0` keeps every round.
Extrude each slice	off	Stack and X-Ray: each sketch's profile is extruded to the next slice and joined into one body, a stepped solid. Use this instead of Loft, which usually fails between two complex profiles. With Outline only ticked, only the outer shape is extruded. Without it, inner loops are filled as well. At 0.5 mm spacing the steps are hard to see.

Setting	Default	What it does
Add 2 rails (90° sections)	off	Stack and X-Ray: two cuts along the stack through the mesh centre, drawn as splines on their own planes and pinned onto every slice's outline (see section 5, *Rails and loft*). The rail planes are built from a line on the picked plane, so they work in parametric designs too.
Loft profiles + rails (new body)	off	Shown with rails. Lofts one new body through the slices along the complete rails, best effort, instead of Extrude each slice.
Hide mesh after	off	Hides the mesh body when it is done, so you can see the result.

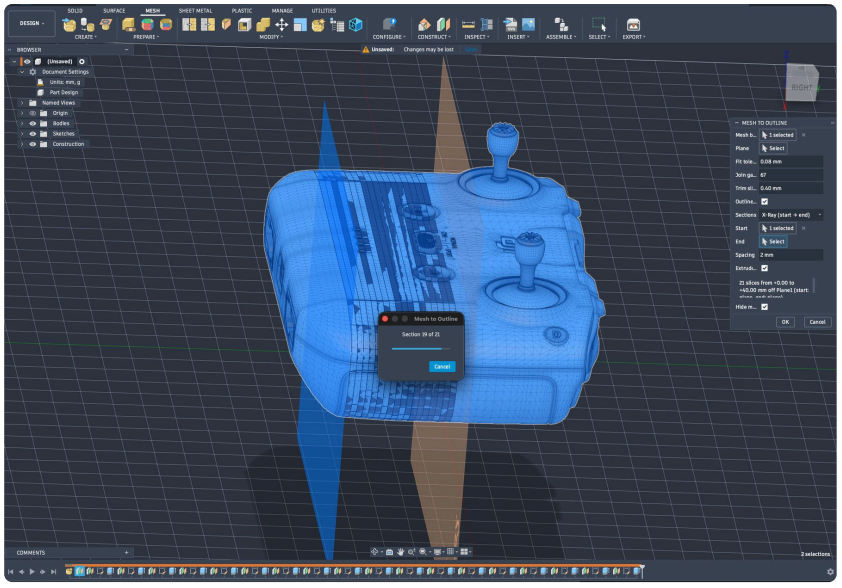
7.4 Walk-through: from mesh to solid with X-Ray

These are real runs on a scanned controller. The setup shot is 10 slices, 4 mm apart. The other shots are 21 slices, 2 mm apart, extruded into one body.

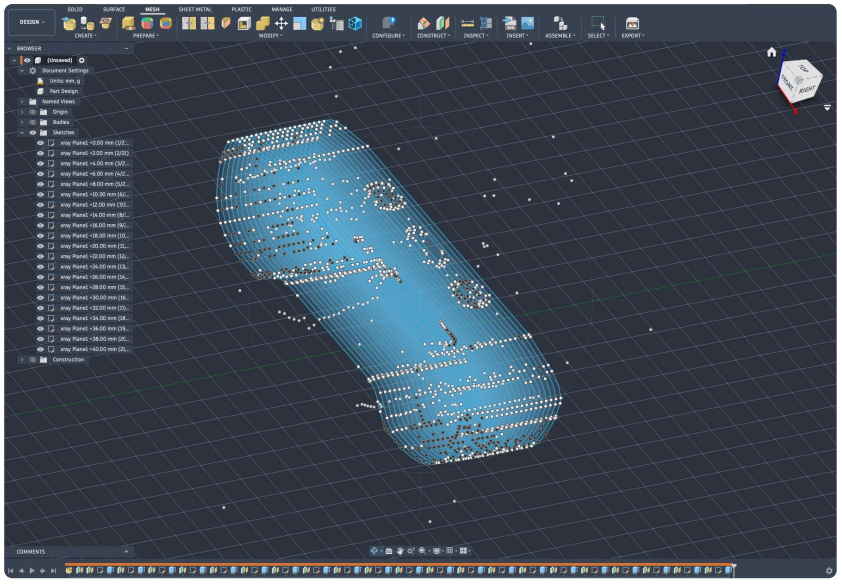
1. Pick two planes. Set **Action** to **X-Ray (start → end)**, choose the mesh body, and pick a Start and an End. Set the **Spacing**. The info line says how many slices that makes (here 10, from +0.00 to +35.00 mm). Tick **Extrude each slice** if you want a solid, then click **OK**.



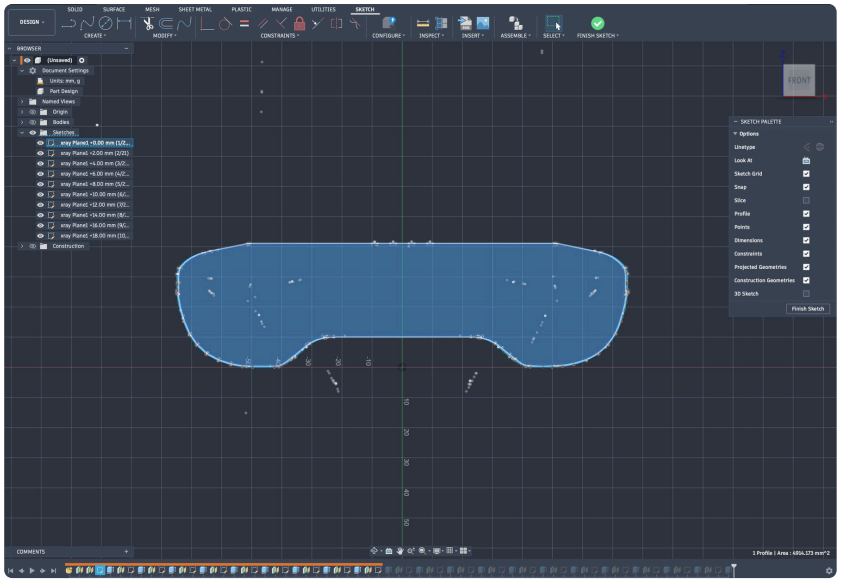
2. Watch it trace. Each slice is cut and fitted in turn. A progress dialog counts the sections. **Cancel** stops it after the current one.



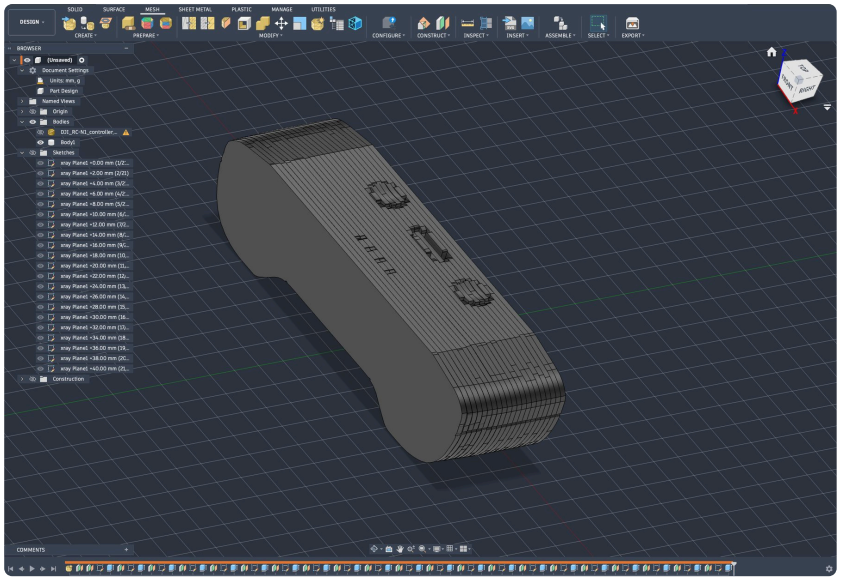
3. One sketch per slice. Every slice lands as its own sketch on its own plane, listed in the browser from first to last and named like `xray Plane1 +4.00 mm (3/21)`. When it finishes, a message box lists each sketch with what it holds (outlines, open curves, lines, arcs, circles, splines, gaps joined, worst miss).



4. A closed profile, every time. Open any sketch: it is one closed outline that Fusion sees as a profile, ready to extrude, offset or dimension.



5. A solid you can design on. With **Extrude each slice** ticked, the stack becomes one stepped solid body in the timeline, next to the mesh.



A single section takes a second or two on a 300,000-triangle mesh. The sketches go in the root component. A mesh inside a component is read in root coordinates, so the curves land on the mesh either way.

7.5 Messages you may see

Message	What to do
Empty cut skipped	The plane misses the mesh there. Fusion's own section sketch would be empty too.
WARNING: the outline did not close in N slices	The mesh has holes wider than **Join gaps** at those slices (a vent grille's slots, say). Those outlines stay open, Outline only drops them, and those slices get no slab, which leaves a groove in the solid. Raise Join gaps to the value the warning gives and run again.
WARNING: N slabs could not be extruded	The sketch was drawn, but Fusion found no profile to extrude there. The per-slice list says why.

Message	What to do
numpy could not be imported / Fusion's Python is now X.Y	Fusion updated its Python. Install the matching add-in version.

7.6 STL to Solid add-in (not released yet)

> This add-in is not available to download yet. Only Mesh to Outline is.

A second add-in, **STL to Solid**, will sit next to Mesh to Outline. It does the face-group conversion (section 3.1, route 2) inside Fusion: it fits planes, cylinders, cones, spheres and tori to a mesh body, builds each one slightly oversized as a surface, and lets Fusion's Boundary Fill keep the material. It is an early preview. It works on clean, CAD-like meshes, and refuses open shells and scans.

Setting	Default	What it does
Mesh bodies	—	The mesh bodies to convert.
Fit tolerance	0.08 mm	How far a triangle may sit off a fitted surface.
Expand surfaces	1.0 mm	How far each surface is grown past its region, so neighbours overlap for Boundary Fill.
Max regions	200	Bodies with more fitted surfaces than this are refused.
Keep tool surfaces	off	Keeps the surface bodies (hidden) after the fill, for checking.
Run Boundary Fill	on	Untick to only create the surfaces and run Boundary Fill yourself.
Hide mesh after	on	Hides the mesh body when a solid was made.
Diagnose rejected tools (slow)	off	If Fusion refuses the fill, finds which surface it rejects.

If the log says *Fusion could not compute the cells*, try Expand 0.5 or 2 mm, or use Mesh → Solid in the app instead.

8. Reading the report

After a Mesh → Solid or Sliced Loft run, the rail scrolls to the report.

PRISMATIC SOLID

Clean BREP with true planes and cylinders — every gate below passed.

CHECK	MEASURED	LIMIT	
Surface deviation p95 (mesh → solid)	0.019 mm	≤ 0.25 mm	PASS
Surface deviation max (mesh → solid)	0.060 mm	≤ 0.26 mm	PASS
Surface deviation p95 (solid → mesh)	0.018 mm	≤ 0.25 mm	PASS
Surface deviation max (solid → mesh)	0.058 mm	≤ 0.26 mm	PASS
Bore deviation p95 (5 bores)	0.015 mm	≤ 0.1 mm	PASS
Volume error	0.08 %	≤ 2 %	PASS

OUTPUT STEP

1,644 triangles → 23 faces

BREP faces	23
Surface types	15 planes, 8 cylinders
Mean deviation	0.005 mm
Worst point at	(−60.91, −3.57, 1.97) mm
Solid volume	3,688 mm³
File size	85.4 KB

Download STEP

Download CadQuery script (.py)

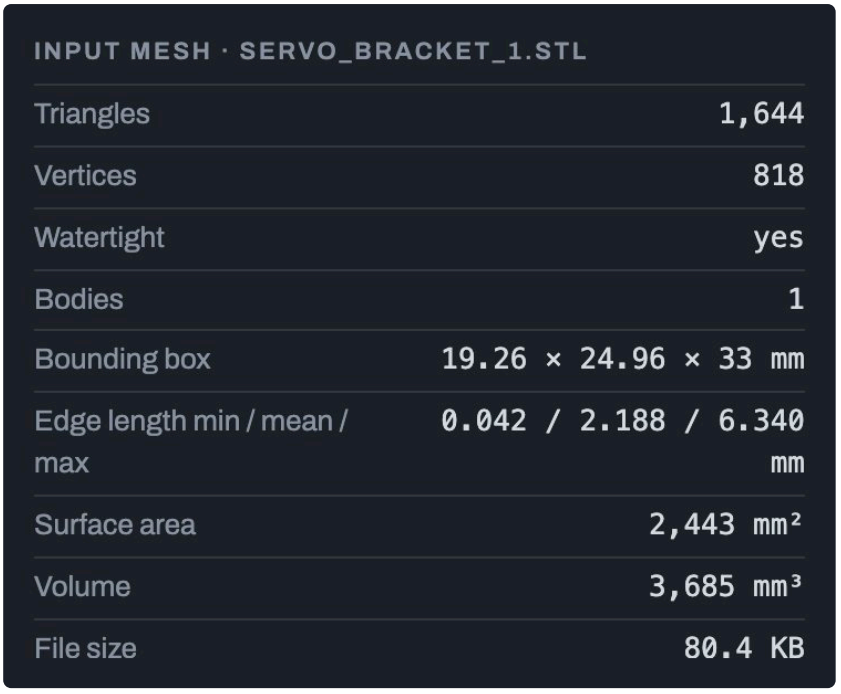
Download Fusion 360 script (.py) — experimental

The script rebuilds the recognised sketches and extrudes as an editable program — change a radius or height and re-run it to get a new STEP.



The result badge, the checks table and the output summary for the servo bracket.

Input mesh. The file as read: triangles, vertices, watertight or not, bodies, bounding box, edge lengths, surface area, volume and file size.



Result badge. Which route made the solid:

Badge	Meaning
Prismatic solid	A clean extrusion with true planes and cylinders. Every check passed. Scripts are offered.
Face-group solid	Surface regions fitted with real planes, cylinders, cones, spheres and tori.
Sliced loft solid	The sliced loft. Checks are shown for information.
Faceted solid	No clean fit passed. The solid is exact and closed, but made of facets.

Bodies table (multi-body files). One row per body: its triangle count, which route it took and details.

Checks table. Each check with the measured value, the limit and **PASS** or **FAIL**. Deviation is measured both ways: from the mesh to the solid and from the solid to the mesh.

Output STEP. The headline shows the reduction, for example `1,644 triangles → 23 faces`. Below it: face count, surface types, mean deviation, where the worst point is, the solid's volume and the file size.

Surfaces, not solids. If the bodies you picked were not closed, the STEP holds surfaces instead of a solid. You can measure them and snap to them, but you cannot cut or join them.

9. Using the results in Fusion 360

Importing a STEP

In Fusion 360 choose **File** → **Open** → **Open from my computer**, or drag the `.step` file into the Data Panel. The part arrives as a solid body with real faces you can sketch on, measure and constrain.

Running a downloaded script

All the Fusion scripts (Mesh → Solid, Sliced Loft, X-Ray and Blueprint) run the same way:

1. Put the `.py` file in an **empty folder of its own**.
2. In Fusion 360: **Utilities** → **Add-Ins** → **Scripts and Add-Ins**.
3. Click **+**, choose that folder, then click **Run**.

For Blueprint, run it in a **new parametric design**. Longer scripts show a progress dialog with a Cancel button. For more detail, open **View** → **Show Text Commands**.

10. Troubleshooting

Problem	What to do
The part is 10× (or 25.4×) too big or small	Change **The file's numbers are in**, or use **…then multiplied by** (for example `0.1`). Watch **Longest side as converted**.
I got a **Faceted solid** instead of a clean one	Open the checks table to see which limit failed. Try a slightly larger **Fit tolerance** or **Surface deviation (max)**. For smooth organic shapes, try **Sliced Loft**.
Conversion runs out of memory (*oom*)	Tick fewer bodies. Each body converted at the same time can use several gigabytes.
Conversion *crashed*	The geometry kernel gave up on one face. Pick fewer bodies, raise the tolerance, or try Sliced Loft.
The result is surfaces, not a solid	The picked bodies are not closed (marked **open**). Pick closed bodies, or repair the mesh first.
A slice outline is broken into pieces	The mesh is leaky. Raise **Join gaps up to**.
Thin slivers or hairpins in the outline	Raise **Trim slivers up to** a little, but keep it below the part's thinnest real feature.
A thin rib or tab vanished from a slice	**Trim slivers up to** is too high. Lower it, or set it to `0` on a watertight mesh.
X-Ray says too many slices or over the time limit	Raise the spacing, narrow the start and end planes, or tick **Outline only**.
Sliced Loft turns a sideways hole into a trough	That is a limit of slicing. Use Mesh → Solid for machined parts.
Blueprint says a key is needed	Paste an API key for the chosen provider, or pick a local *Custom* server.
Blueprint misread a number	Fix it in the **Parameters** table and click **Rebuild**. No new read is needed.
Mesh to Outline left a groove in the extruded solid	Some slices did not close. Raise **Join gaps up to** to the value the warning gives.
A red banner appears at the top	Read the message. Open **details** for more, then dismiss it with **×**. Your file and settings stay loaded.

11. Quick reference

File types

Kind	Extensions	Opens in
Mesh	`.stl` `.obj` `.ply` `.off` `.3mf` `.glb` `.gltf`	Mesh → Solid, Sliced Loft, X-Ray
Drawing	`.jpg` `.png` `.webp` (or paste from the clipboard)	Blueprint

Default settings

Setting	Default
Units / multiplier	mm / 1
Fit tolerance	0.08 mm
Surface deviation p95 / max	0.25 mm / 0.26 mm
Bore deviation (max)	0.10 mm
Volume error	2 %
Faceted reduce tolerance	0.05 mm
Slice spacing (Sliced Loft, X-Ray, Mesh to Outline)	0.2 mm
Join gaps up to	2.5 mm
Trim slivers up to	0 mm
Tidy corners (X-Ray, Mesh to Outline)	off; Sharpen rounds below 0.08 mm when on
Rails / loft (X-Ray, Mesh to Outline)	off; the loft is skipped past 60 slices
X-Ray limits	500 slices, about 4 minutes of fitting per download
Mesh to Outline stack	10 sections (2 to 400)
Blueprint effort	high

What each tool gives you

Tool	STEP	Fusion 360	Other
Mesh → Solid	Yes	Script: parametric timeline (prismatic) or Boundary Fill (face-group)	CadQuery script (prismatic)
Sliced Loft	Yes	Script: section sketches + Loft per run	—
X-Ray	—	Script: one sketch per slice, optionally extruded into slabs	—
Blueprint	Yes	Script: User Parameters, sketches, extrudes (recipe mode)	3D preview; four rendered views; the CadQuery script (code mode)
Mesh to Outline (add-in)	—	Sketches drawn straight into the design, optionally extruded into one body	—

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12. Using stlToSolid from an AI assistant

stlToSolid can run as an **MCP server**. MCP (Model Context Protocol) is the way AI assistants such as Claude and Cursor use outside tools. Connect it once, then ask in plain words: *"Convert ~/Downloads/bracket.stl to a STEP and tell me how close it is to the mesh."*

The server runs on your own computer, like the app. Your files are not uploaded by stlToSolid. What the assistant reads, such as the report and the slice pictures, goes to whichever AI service you use, as with anything else you show it. You need no API key here: drawings are read by the assistant you are already talking to.

12.1 Installing

You need uv, a small Python installer. It fetches the right Python by itself.

Step 1: test it in a terminal. The first run downloads the geometry kernel, about 2.8 GB, and takes a few minutes. Many assistants give up on a server that takes this long to start, so do this first. It also tells you whether everything works.

```
uvx --python 3.12 --from "stltosolid[mcp]" stltosolid-mcp --check
```

Step 2: connect your assistant.

Claude Code:

```
claude mcp add stltosolid -- uvx --python 3.12 --from "stltosolid[mcp]" stltosolid-mcp
```

Claude Desktop, Cursor and others: add this to the client's MCP settings. In Claude Desktop that is **Settings** → **Developer** → **Edit Config**. Restart the client afterwards.

```
json { "mcpServers": { "stltosolid": { "command": "uvx", "args": ["--python", "3.12", "--from", "stltosolid[mcp]", "stltosolid-mcp"] } } }
```

Claude Desktop, one click: download `stltosolid-<version>.mcpb` from the Releases page and open it. It asks where results should go.

With the desktop app installed: the app contains the server, so no download is needed. Point the client at the app with the `--mcp` flag.

On a Mac:

```
/Applications/stlToSolid.app/Contents/MacOS/stlToSolid --mcp
```

On Windows:

```
C:\path\to\stlToSolid\stlToSolid.exe --mcp
```

12.2 What the assistant can do

Tool	What it does
`inspect_mesh`	Size, triangle count, bodies, a warning if the units look wrong, and a suggested method.
`convert_mesh`	Mesh → Solid or Sliced Loft, with every setting from the app. Gives a STEP, and Fusion and CadQuery scripts where they exist.
`job_status`, `cancel_job`	Follow or stop a long conversion.
`suggest_loft_axis`	Which axis a loft would slice along.
`trace_section`	One slice, as a picture the assistant can see, with the fitted curves counted.
`plan_xray`, `xray_fusion_script`	X-Ray: where the planes fall, and the Fusion script of real sketches.
`inspect_step`	Reads a STEP back the way CAD will: solids, faces, surface types, volume, closed or not.
`preview_recipe`, `build_recipe`	Blueprint: draws a recipe from four sides, then builds the STEP and parametric Fusion script.
`build_cadquery`	Blueprint's code mode, for bends and blends. **Off by default**, see below.
`fusion_handoff`	The Python that puts a result into Fusion through Autodesk's own MCP server.

There are also four ready-made prompts your client may show in its menu: `convert-for-fusion`, `read-drawing`, `xray-to-fusion` and `send-to-fusion`.

12.3 Long conversions

A big part can take many minutes. The assistant gets an answer after 45 seconds with a job number, and keeps asking until the job is done. The work carries on in the meantime, one conversion at a time. If you quit the assistant, its conversion stops too, so nothing keeps running unseen.

12.4 Where files go

Results go in one folder per part under `~/stlToSolid/mcp-output`. Change it with `--out` or `STLTOSOLID_OUT`. A file is never overwritten: a second run writes `part-2.step`.

To limit which files the server may read, start it with `--root <folder>`.

12.5 Allowing the code tool

`build_cadquery` runs a CadQuery script the assistant writes. It handles shapes the recipe cannot draw. It is off unless you start the server with `--allow-code`, or set `STLTOSOLID_MCP_ALLOW_CODE=1`, or tick the box in the Claude Desktop bundle.

Scripts may only use `cadquery` and `math`, have no file access, run in their own process and stop after two minutes. Turn it on only with an assistant you trust.

12.6 Sending results straight into Fusion

Autodesk has its own Fusion MCP server. Turn it on in Fusion under **Preferences** → **General** → **API**, and connect it to the same assistant. Then ask: "*Put that part in my open design.*" The assistant uses `fusion_handoff` here and Autodesk's script tool there. Fusion can undo the result.

12.7 Running it on a NAS or server

The Docker image contains the server too, off by default. Choose a long random token and start it:

`STLTOSOLID_MCP_TOKEN=your-long-random-token STLTOSOLID_MCP_URL=http://your-nas:8322 docker compose --profile mcp up -d`

Clients connect to `http://your-nas:8322/mcp` and send the token as a bearer token. A remote server cannot see your disk, so files live in the folder `data/mcp` on the server:

- **Get a file in:** copy it into a share of that folder, or send it: `curl -T part.stl -H "Authorization: Bearer your-long-random-token" http://your-nas:8322/files/part.stl`
- **Tell the assistant** the path it reports, such as `uploads/part.stl`.
- **Get results out:** every result comes with a download link.

Use this only on a network you trust, because the connection is not encrypted. Put an HTTPS proxy in front of it for anything wider.

12.8 If it does not work

Problem	What to do
The assistant says the server failed to start	Run the `--check` command in a terminal. The first download takes minutes, and clients give up sooner.
`uvx: command not found`	Install uv. If it is installed and the client still cannot find it, use the full path to `uvx` in the settings.
A path is refused	The server needs absolute paths. On Windows that means `C:\Users\...`. Check `--root` if you set one.
"Still running" for a long time	That is normal for a large part. Ask the assistant to keep checking, or to cancel and convert fewer bodies.
It ran out of memory	Convert fewer bodies at once: the `bodies` setting takes the numbers from `inspect_mesh`.